Possible Topics for Literature Review Final Exam Presentation and Paper

You should pick a topic from the following list (or propose something to me). It cannot be closely related to your research and must be approved by me. E-mail me a proposed topic with a list of resources (2023 NSAC Long Range Plan in Nuclear Science, journal articles (*Physical Review C, Physics Letters B, Nuclear Physics A,* etc.) you've found by next Friday 10/31 at 5PM.

Presentations should be self-contained and provide an overview of a topic but need not be a full review.

- Islands of Inversion
- **Shape Coexistence – Aashish**
- Non-axial deformation
- The N=126 Magic Number
- **Superheavy Nuclei – Noah (Z=126)**
- Fusion Reactors
- Reactor Neutrino Anomaly
- Neutrinoless double beta-decay
- Beta-Delayed Particle Emission
- **s-process nucleosynthesis -- Amadie**
- r-process nucleosynthesis
- rp-process nucleosynthesis
- **Nuclear Pasta in Neutron Stars -- Nick**
- **Halo Nuclei -- Mark**
- **Nuclei and Medical Imaging -- Alan**
- Radiation Therapy Techniques
- **Medical Isotope Production -- Unish**
- Effective Field Theory
- Rare Isotope Harvesting
- **Neutron Imaging -- Sushma**
- Astromers
- Nuclear Masses and Astrophysics
- Nuclear Materials Analysis
- **Density Functional Theories- -- Parker**
- Alpha Clustering in Nuclei